

Total No. of Questions : 8]

SEAT No. :

PE-5969

[Total No. of Pages : 3

[6582]-29

S.E. (Electronics/ET&C/Electronics (VLSI

Design & Tech.)/E.C (A.C.T.))

DATA STRUCTURES

(2019 Pattern) (Semester - III) (204184)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable additional data if necessary.
- 5) Use of Calculator is allowed.

- Q1)** a) Write a short note on circular queue. Compare it with linear queue. [5]
b) Convert the following prefix expression into infix form. Show all the steps and stack contents: [6]

*-A/BC-/AKL

- c) Write ADD and DETETE function in 'C' for Queue using array. [6]

OR

- Q2)** a) Compare Stack and Queue. What are the advantages of circular queue over liner queue? [6]
b) Write a function PUSH and POP in 'C' for stack using linked list. [6]
c) What are the applications of Queue? Explain two applications in detail. [5]

- Q3)** a) Draw and explain circular linked list. State the limitations of single linked list. [5]
b) Write a 'C' function to insert a number at end in to the singly linked list. [6]
c) Differentiate singly linked list and doubly linked list. [6]

OR

- Q4)** a) Compare array and linked list. [5]
b) Write a 'C' function to delete a number from singly linked list. [6]
c) Explain doubly linked list (DLL). What are the advantages of DLL over SLL. [6]

P.T.O.

Q5) a) Define BST? Create a BST for the following data: [6]
14, 15, 4, 9, 7, 18, 3, 5, 7.

b) Define binary tree. Name and explain with suitable example the following terms : [6]

- i) Root node
- ii) Left sub tree and right sub tree
- iii) Depth of tree

c) Construct the binary search tree from the following elements: [6]
15, 4, 16, 8, 2, 18, 14

Also show preorder, inorder and postorder traversal for the same.

OR

Q6) a) Construct Binary search tree for the following : [6]
MAR, OCT, JAN, APR, NOV, FEB, MAY, DEC, JUN, AUG, JUL, SEP

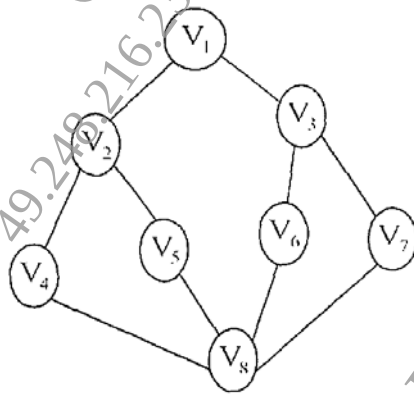
b) Write a pseudo code to search an element in binary search tree using arrays. [6]

c) Explain with suitable example how binary tree can be represented using:

- i) Array
- ii) Linked List

[6]

Q7) a) Draw adjacency list and adjacency matrix for the following graph: [6]

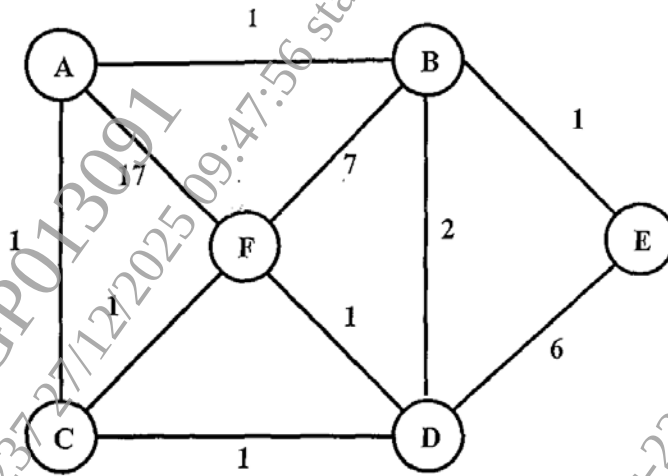


b) What is MST? Explain with suitable example Kruskal's Algorithm to find out. MST [6]

c) Define DFS and BFS graph with example. [6]

OR

- Q8) a)** Explain Kruskal algorithm? Find the minimum spanning tree for below figure. Using Kruskal's Algorithm. **[6]**



- b) Explain Dijkstra's algorithm with example. **[6]**
c) Explain with suitable example the techniques to represent a Graph. **[6]**
Note: consider graph of minimum 5 vertices